Conclusion

Conclusion I

We have presented a configurable, reproducible meta-analysis template that turns a single search term into a complete, evidence-bound portrait of its literature. Applied to **Modafinil**, it retrieved and de-duplicated 2302 records across 7 engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, and ChinaRxiv), classified them into 6 configurable subfields (with **Clinical Sleep** dominant at 64.3%), extracted 5 topics over a 500-feature vocabulary, computed reproducible document embeddings, mapped the citation network (2204 nodes, 8,772 edges, 1377 communities), and framed 6 hypotheses for optional evidence scoring.

Key Findings I

The analysis answers the four research questions posed in the introduction:

1. **RQ1 (Growth)**: The modafinil literature spans 26 years (2000–2026) and grows at a CAGR of 3.45%, doubling every 11.3 years. The peak year 2025 produced 112 publications, indicating sustained and active research interest.
2. **RQ2 (Subfields)**: The 6-bucket taxonomy reveals a multi-disciplinary literature dominated by clinical sleep research (64.3%), with significant representation from cognition, psychiatry, and pharmacology.
3. **RQ3 (Topics)**: NMF extracted 5 latent topics — cognitive enhancement, ADHD treatment, pharmacological dose-response, sleep disorders, and psychiatric fatigue — that cross-cut the explicit subfield taxonomy and reveal the thematic structure of the field.

Key Findings II

4. **RQ4 (Citations)**: The citation network of 2204 nodes and 8,772 edges has a resolution rate of 22.6%, 1377 communities, and a maximum in-degree of

   4.165 The heavy-tailed degree distribution is characteristic of citation networks, with a small number of foundational works anchoring the structure.

Architectural Contribution I

The contribution is architectural rather than topical: every domain-specific value flows from one configuration file and the pipeline's own outputs into a generated manuscript, so the same machinery re-targets to any topic by editing configuration alone. Combined with an offline, deterministic default run, this yields a *living literature review* — a synthesis that can be re-executed on demand as a field evolves, with every number traceable to a regenerable artifact.

Reproducibility I

This manuscript was generated from a live retrieval run using 7 engines. Every number, table, and figure in this document is injected from a committed artifact (`output/data/*.json`, `output/figures/*.png`). Re-running the pipeline with the same configuration reproduces identical data outputs; the 18 figures are deterministic given fixed seeds, and the manuscript text is regenerated from the same template. No number in this document was typed by hand.